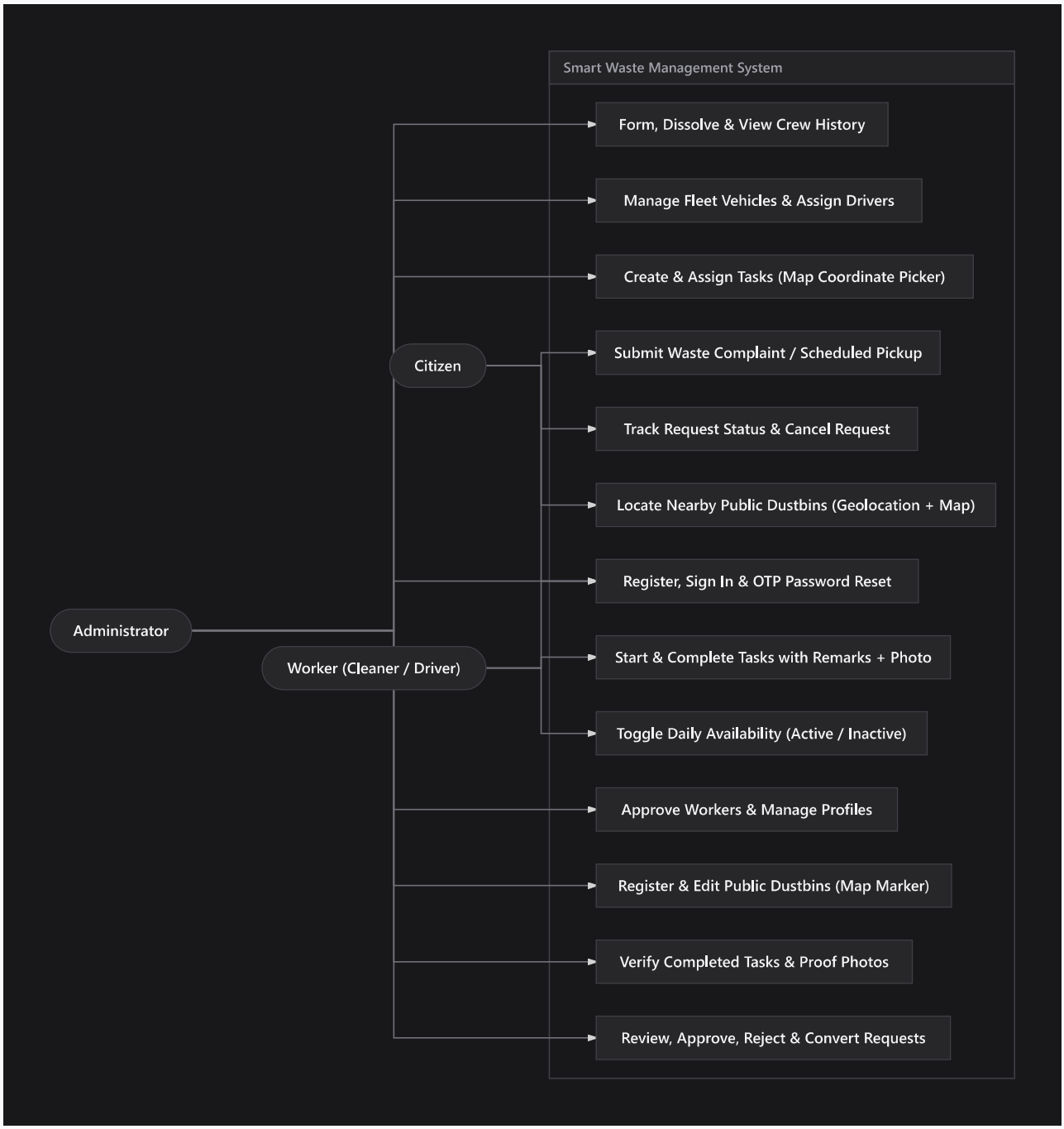
UML Diagrams — Smart Waste Management System (SWMS)

> NOTE
>
> All diagrams below are derived directly from the live Django implementation (`accounts`, `citizen_services`, `workforce`, and `config` apps) and PostgreSQL schema.

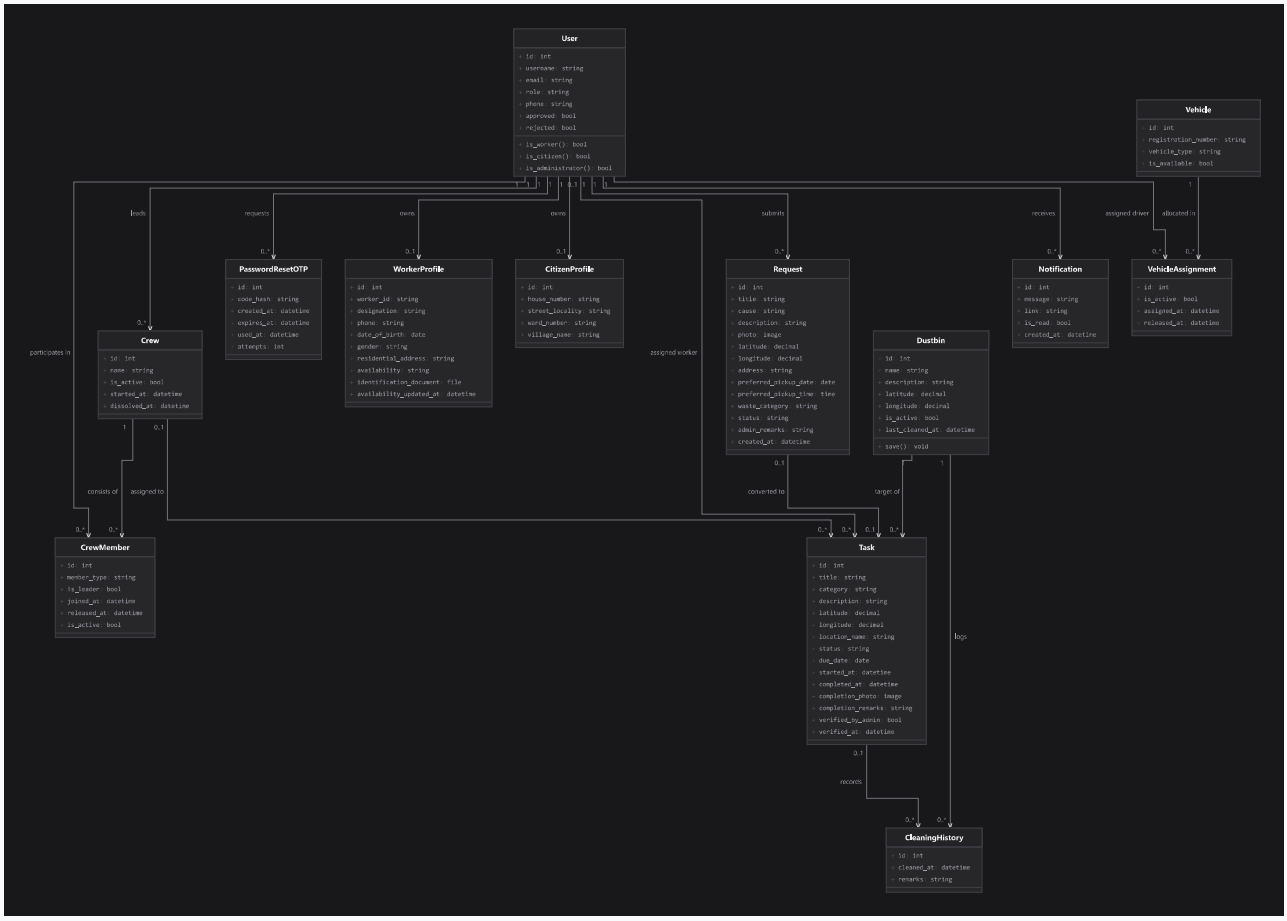
1. Use Case Diagram

Models the three primary actors (**Citizen**, **Administrator**, **Worker**) and their interactions with system capabilities.



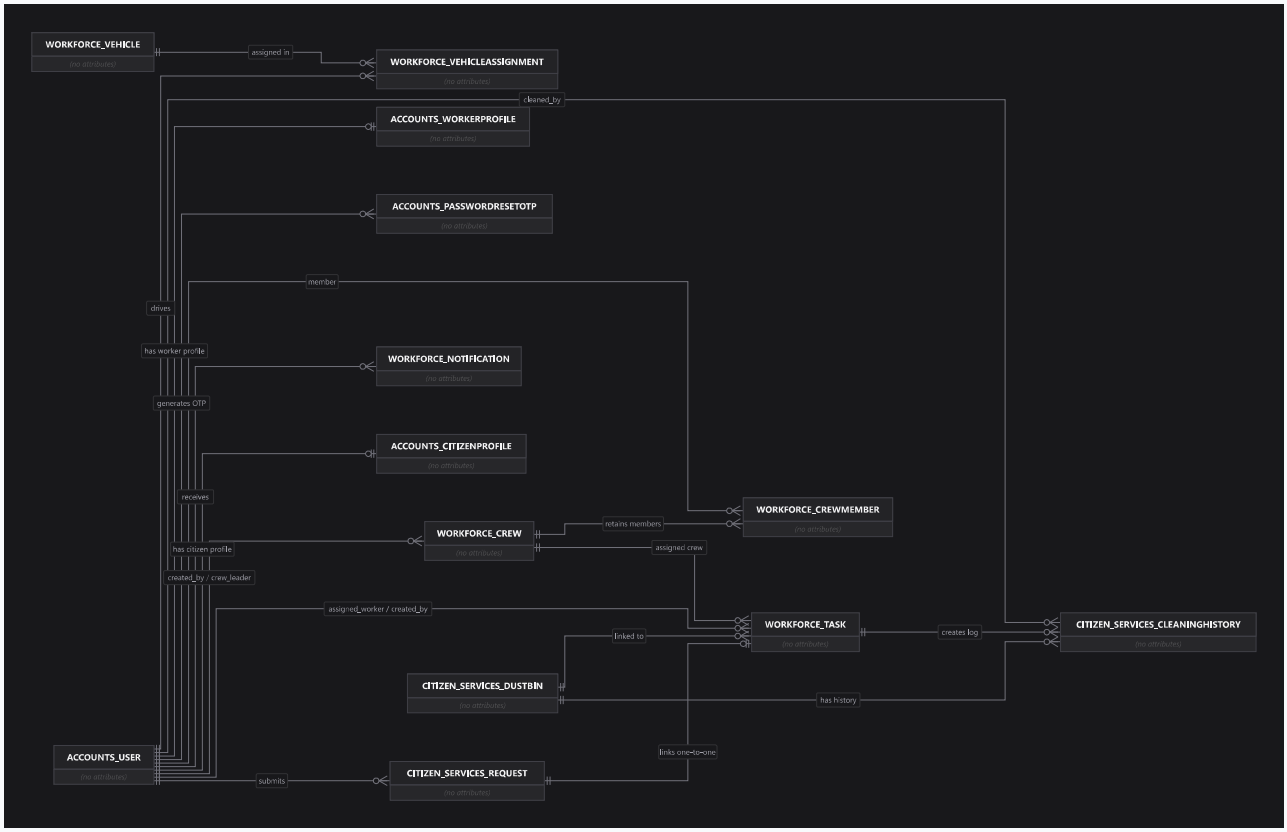
2. Class Diagram

Represents the domain classes across `accounts`, `citizen_services`, and `workforce` along with their key attributes, methods, and associations.



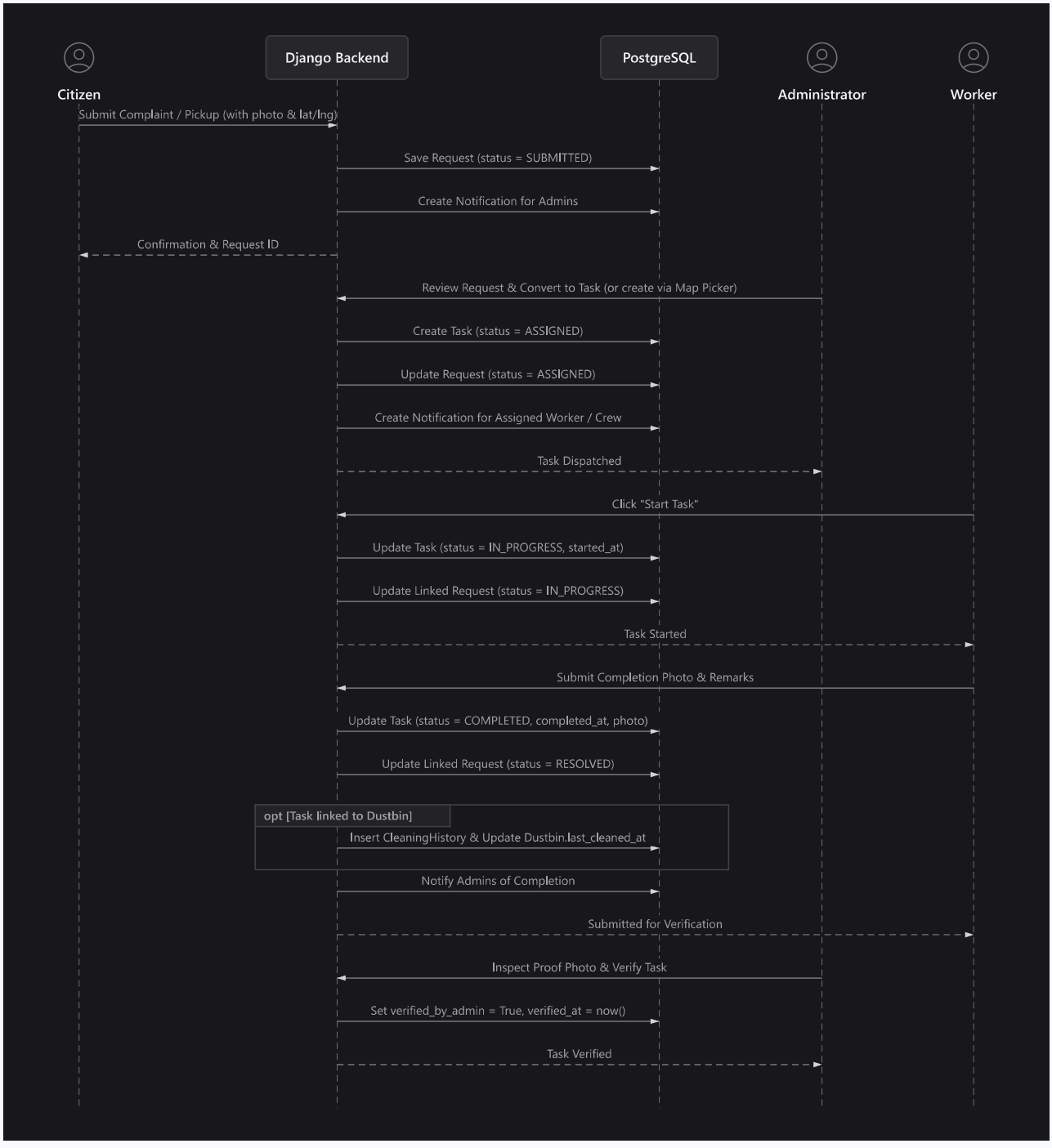
3. Entity-Relationship (ER) Diagram

Illustrates the PostgreSQL relational database tables and foreign key cardinalities.

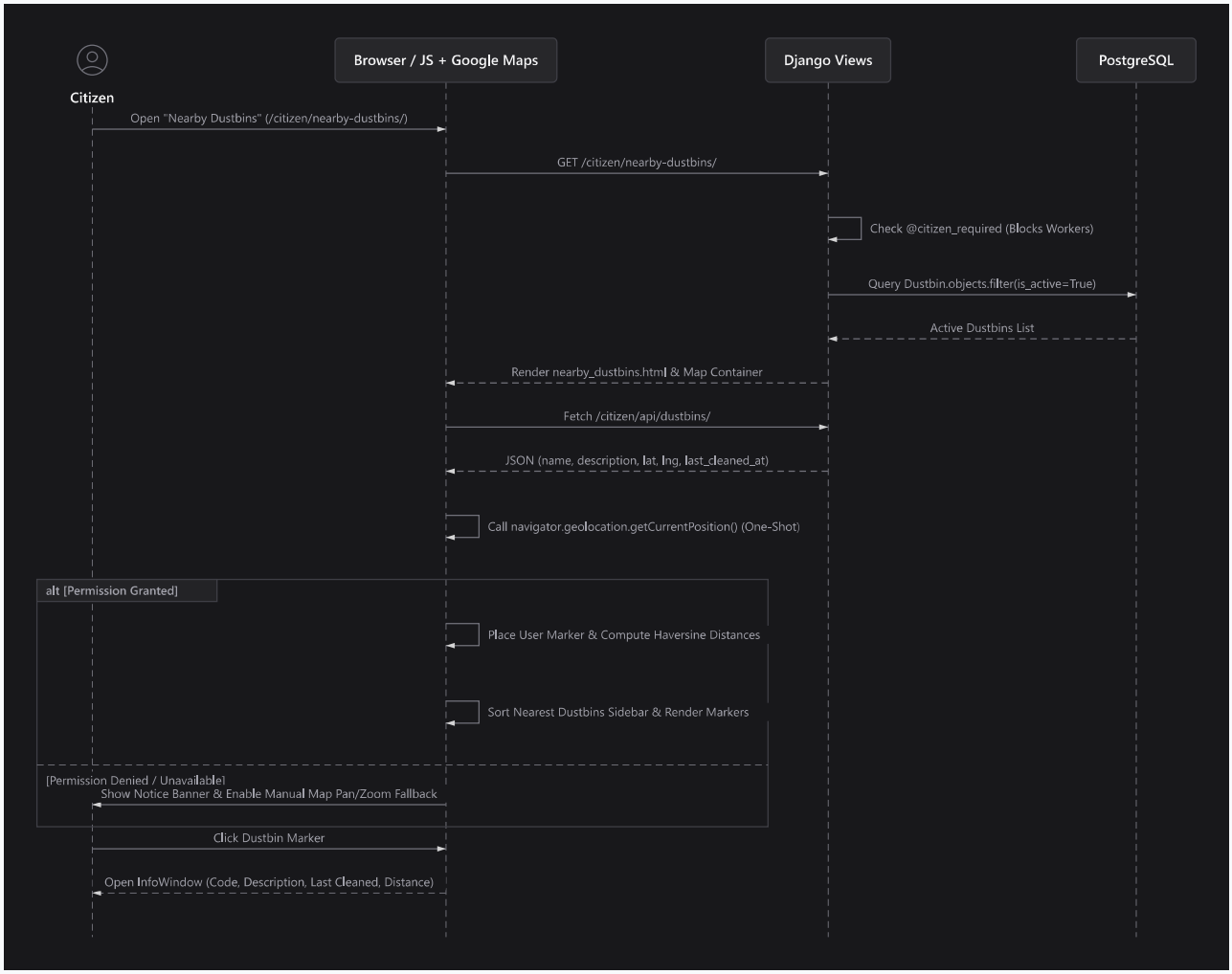

WORKFORCE_VEHICLE
(no attributes)
assigned in
WORKFORCE_VEHICLEASSIGNMENT
(no attributes)
cleaned_by
ACCOUNTS_WORKERPROFILE
(no attributes)
ACCOUNTS_PASSWORDRESETOTP
(no attributes)
member
drives
has worker profile
WORKFORCE_NOTIFICATION
(no attributes)
generates OTP
ACCOUNTS_CITIZENPROFILE
(no attributes)
receives
WORKFORCE_CREWMEMBER
(no attributes)
WORKFORCE_CREW
(no attributes)
retains members
assigned crew
has citizen profile
created_by / crew_leader
assigned_worker / created_by
WORKFORCE_TASK
(no attributes)
creates log
CITIZEN_SERVICES_CLEANINGHISTORY
(no attributes)
linked to
CITIZEN_SERVICES_DUSTBIN
(no attributes)
has history
links one-to-one
ACCOUNTS_USER
(no attributes)
submits
CITIZEN_SERVICES_REQUEST
(no attributes)

4. Sequence Diagrams

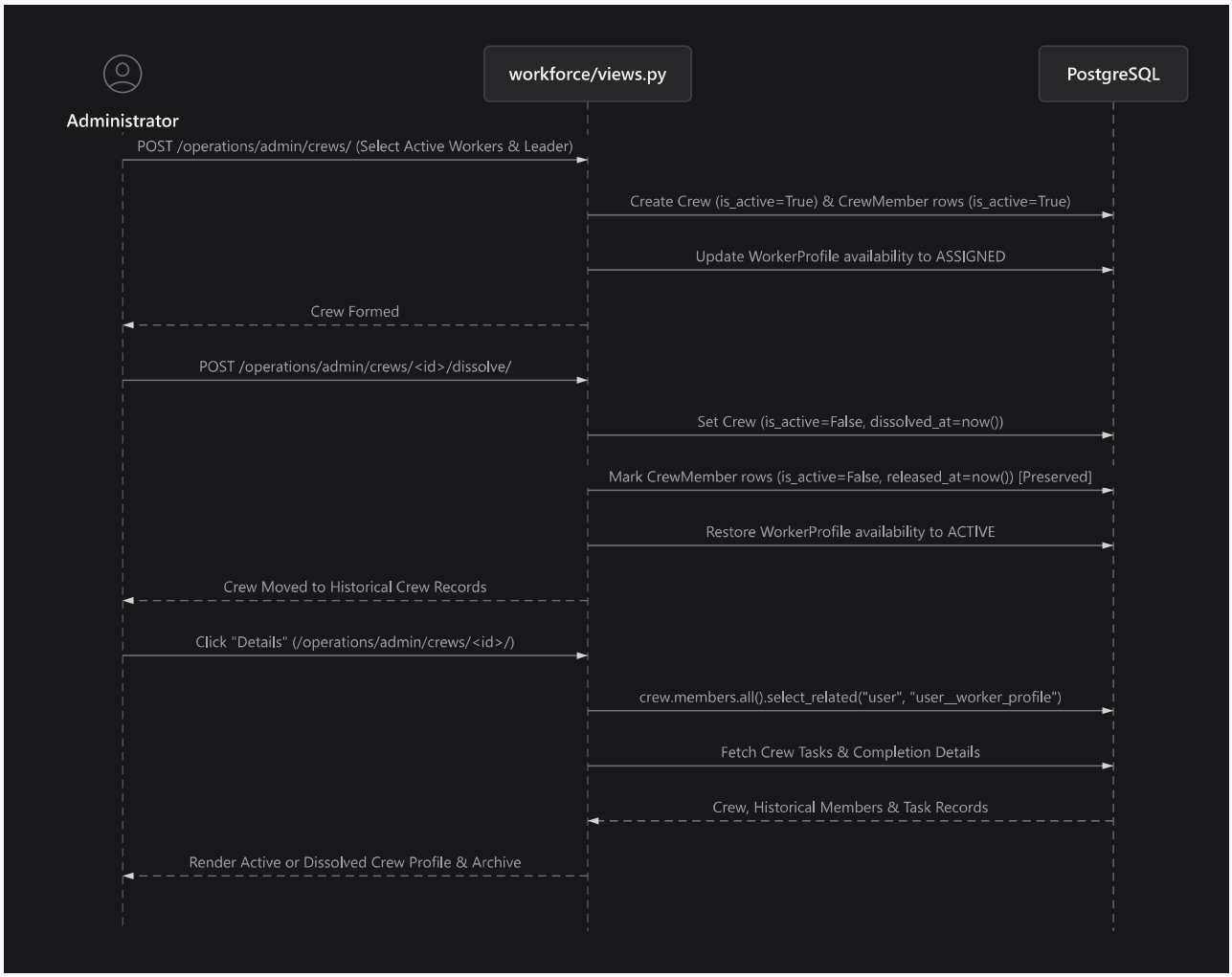
4.1 End-to-End Request Submission, Task Dispatch, Execution & Verification



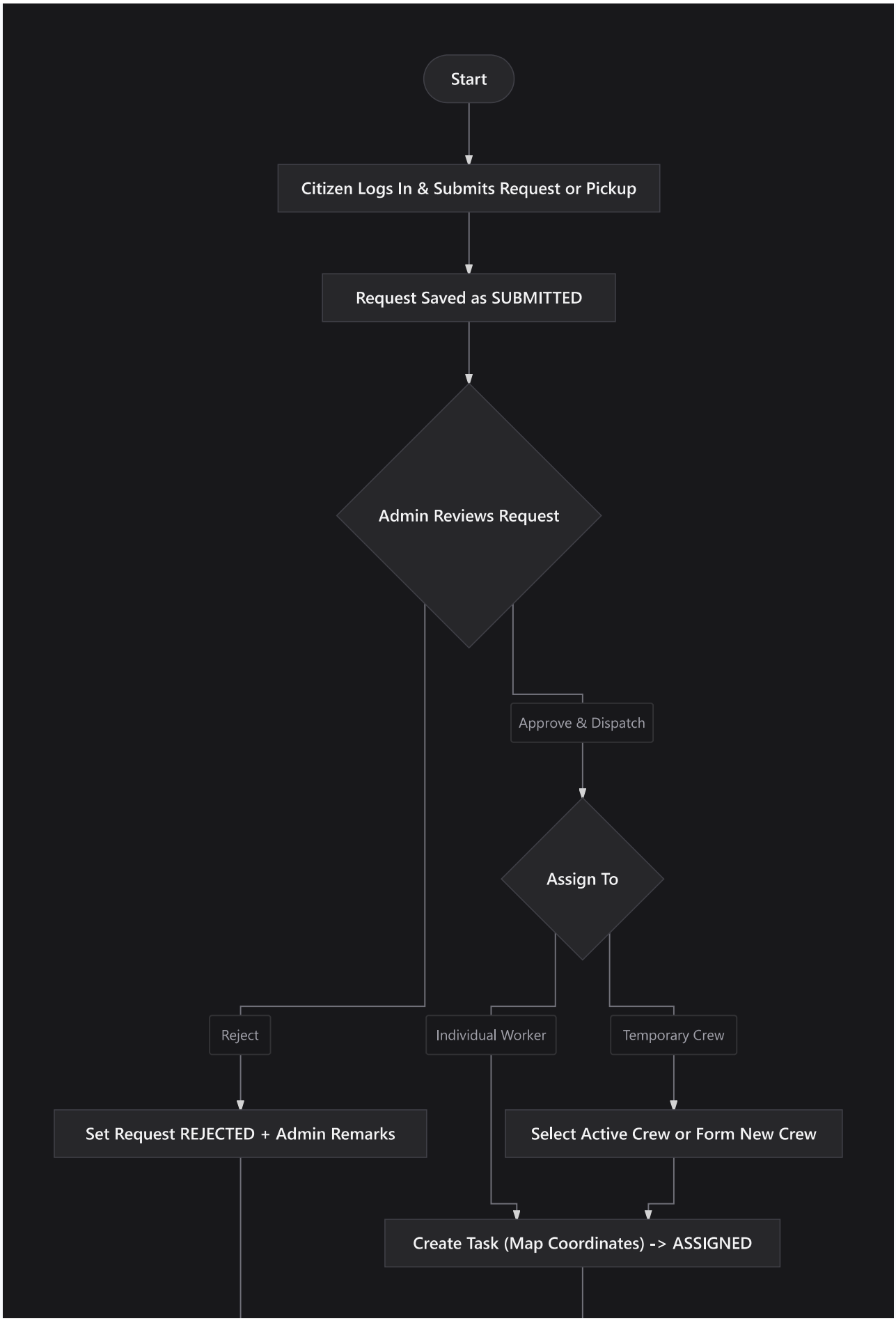
4.2 Citizen Nearby Dustbins Discovery (One-Shot Geolocation)

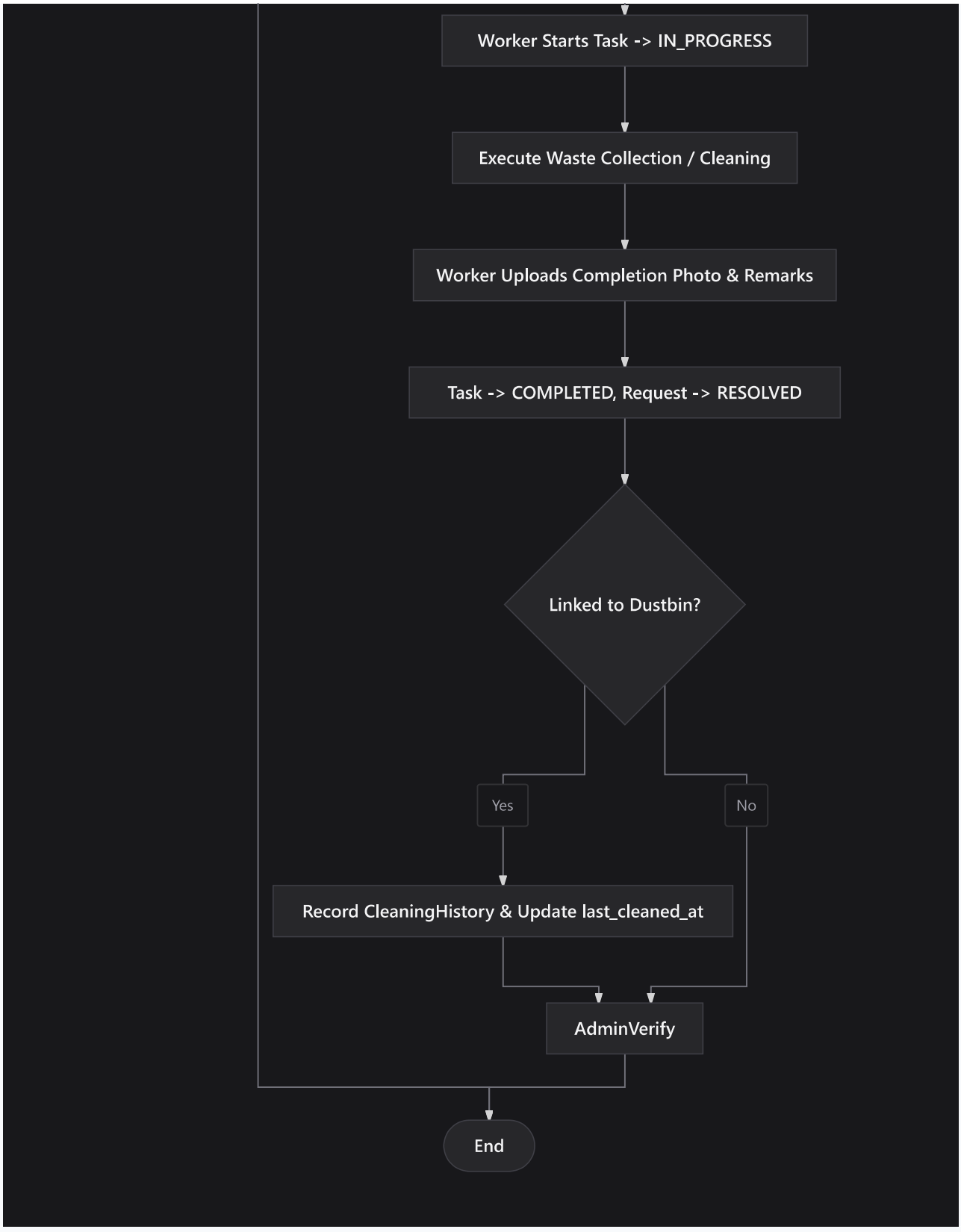


4.3 Crew Formation, Dissolution & Historical Archive Access



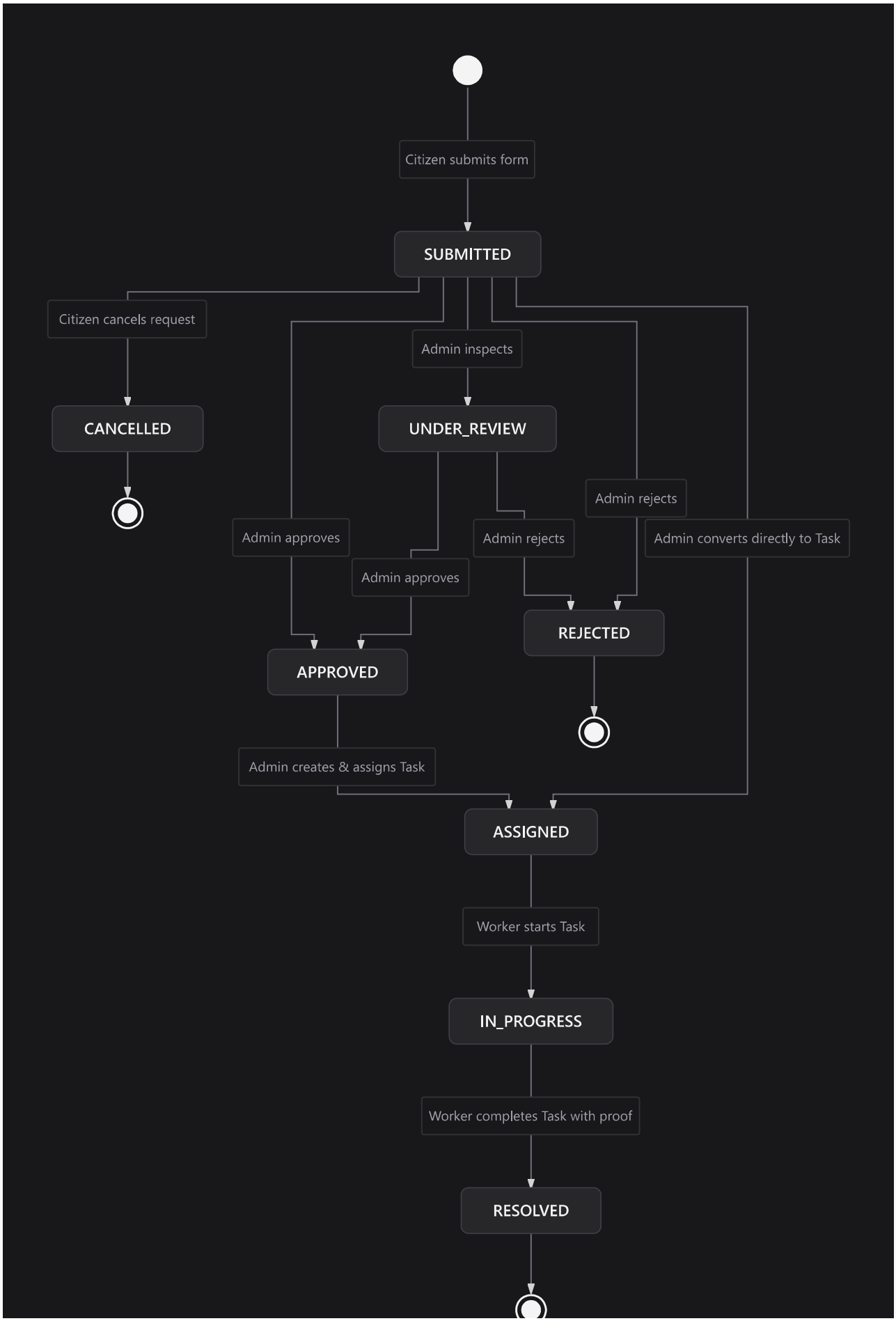
5. Activity Diagram (Core Municipal Waste Workflow)


Start
Citizen Logs In & Submits Request or Pickup
Request Saved as SUBMITTED
Admin Reviews Request
Approve & Dispatch
Assign To
Reject
Individual Worker
Temporary Crew
Set Request REJECTED + Admin Remarks
Select Active Crew or Form New Crew
Create Task (Map Coordinates) -> ASSIGNED


Worker Starts Task -> IN_PROGRESS
Execute Waste Collection / Cleaning
Worker Uploads Completion Photo & Remarks
Task -> COMPLETED, Request -> RESOLVED
Linked to Dustbin?
Yes
No
Record CleaningHistory & Update last_cleaned_at
AdminVerify
End

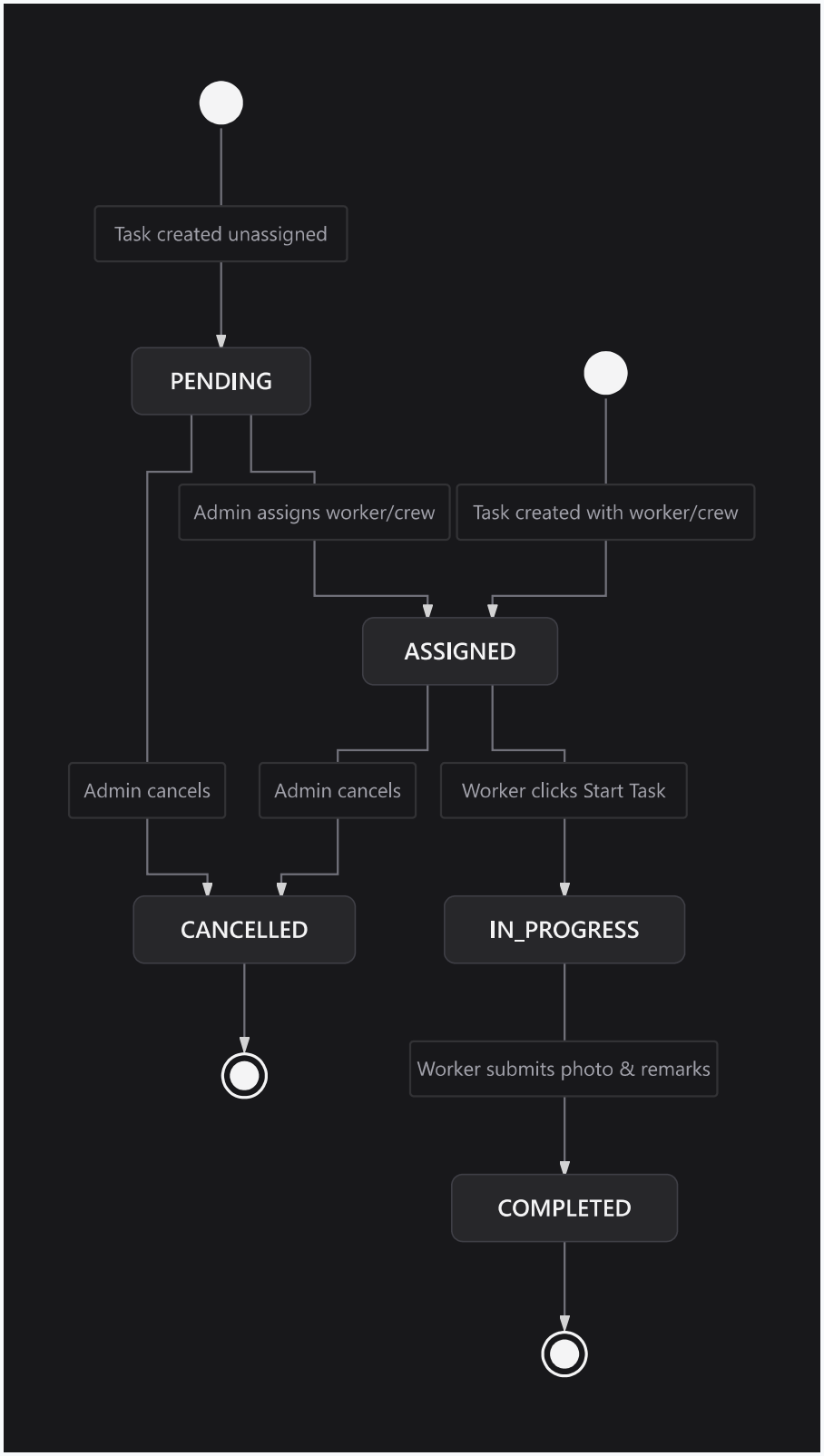
6. State Transition Diagrams

6.1 Citizen `Request.Status` Lifecycle


Citizen submits form
SUBMITTED
Citizen cancels request
Admin inspects
CANCELLED
UNDER_REVIEW
Admin rejects
Admin approves
Admin rejects
Admin converts directly to Task
Admin approves
REJECTED
APPROVED
Admin creates & assigns Task
ASSIGNED
Worker starts Task
IN_PROGRESS
Worker completes Task with proof
RESOLVED



6.2 Operational `Task.Status` Lifecycle



7. System Architecture / Component Diagram

